

Dusts/Other Mine Aerosols

Chapter 4



When the Scrubber Is Turned on . . .



**What's like fter the scrubber has been on
for only a few minutes . . .**



Aerosol Types/Definitions

Aerosol – any mass of solid or liquid particles suspended in a gas

Dust – solid particulate matter in a gas; most common; from fragmentation or resuspension; 1-100 μm diameter; generally 1-20 μm ; particles above 20 μm relatively quickly settle

Aerosol Types/Definitions

Fumes – solid products from combustion, sublimation or distillation; usually $< 1 \mu\text{m}$; note especially diesel particulate matter

Smoke – particles generally 0.01-1 μm ; usually visible; do not result from condensation

Aerosol Types/Definitions

Fog – liquid particles in a gas; condensation of liquid or dispersal of small liquid droplets; usually due to temperature change; few μm to 100 μm ; also mist

Smog – combination of smoke and fog; $< 1 \mu\text{m}$; contains photochemical reaction products and water vapor

Aerosol Types/Definitions

Haze – similar to smog; $< 1 \mu\text{m}$; solid particles and water vapor

Generally accepted size ranges, sampling and control methods for each are given in Fig. 4.1 on pp. 79-80 (accepted by American Society of Heating, Refrigerating & Air Conditioning Engineers)

Dynamic Behavior of Aerosols

Brownian Motion – controlling mechanism for particles $< 0.1 \mu\text{m}$ suspended in quiescent atmosphere; random movement of particles caused by movement of gas molecules that “hit” the particles; resulting mixing of particles and gases called diffusion

Stokes' Law

Used to determine settling velocity of dust particles $> 1 \mu\text{m}$ falling in a quiescent setting with Reynolds numbers < 1

Settling velocity determined by equating drag force on falling (spherical) particle to force of gravity (~10% error)

Stokes' Law

If particles are non-spherical, then Stokes' diameter can be used instead

Stokes' diameter – diameter of hypothetical spherical particle with same density and settling velocity as the non-spherical particle

Stokes' Law – Derived Equation Terminal settling velocity:

$$V_t = \frac{\rho_p D_p^2 g}{18 \mu} \quad (\text{Eq. 4.1})$$

ρ_p - density of particle, kg/m^3

D_p - diameter of particle, m

g - gravity constant, 9.807 m/s^2

μ - viscosity of air, $1.81 \times 10^{-5} \text{ Pa}\cdot\text{s}$

For Reynolds number ≤ 1.0 ; diameter $> 1 \mu\text{m}$

Accompanying equation for Reynolds number check:

$$N_{Re} = \frac{\rho_p V_t D_p}{\mu} \quad (\text{Eq. 4.2})$$

V_t – relative velocity of particle

Note: This is particle Reynolds number, different from Reynolds number for fluid flow (introduced later); if Reynolds number > 1.0, another equation must be used (see Hinds, 1982, p. 51)

- When particle < 1 μm , equation 4.1 is inaccurate (slippage at particle surface)
- Particle falls faster than predicted
- Must apply Cunningham correction factor, C_c , to the velocity calculation

Cunningham correction factor:

$$C_c = 1 + \frac{2.52\lambda}{D_p} \quad (\text{Eq. 4.3})$$

λ – mean free path of molecules of gas, m, which is the average distance a gas molecule travels before colliding with another molecule

For air, $\lambda = 6.6 \times 10^{-8}$ m at 20°C, 1 atm; allows use of Stokes' law for particles as small as 0.1 μm

Include C_c in equation 4.1, when necessary

Behavior of Aerosols in a Moving Airstream

- Dust concentrations downwind of a dust source function of rate of convection, diffusion, agglomeration, and other depositional factors
- Thus, dust concentrations can be hard to predict
- Some studies done in past

Classification of Mineral Dusts and Other Relevant Aerosols

Fibrogenic dusts:

- a. silica (quartz, cristobalite, tridymite, chert)
- b. silicates (asbestos, talc, mica, others)
- c. metal fumes (nearly all)
- d. beryllium, tin, iron ores
- e. carborundum
- f. coal

Carcinogenic aerosols:

- a. asbestos
- b. radon daughters (attached to any dust)
- c. arsenic
- d. diesel particulate matter (DPM)
- suspected
- e. silica – suspected

Toxic (poisonous) aerosols:

- a. dusts of ores of beryllium, arsenic, lead, uranium, radium, thorium, chromium, vanadium, mercury, cadmium, antimony, selenium, manganese, tungsten, nickel, silver (principally oxides and carbonates)
- b. mists and fumes of organic and other body-sensitizing chemicals

Radioactive dusts:

- a. ores of uranium, radium, and thorium (alpha and beta rays)
- b. dusts with radon daughters attached (alpha radiation)

Explosive dusts (airborne):

- a. metallic dusts (magnesium, aluminum, zinc, tin, iron)
- b. coal (bituminous, lignite)
- c. sulfide ores
- d. organic dusts

Nuisance dusts:

- a. gypsum
- b. kaolin
- c. limestone

Note: can overload lung clearance mechanism, causing some respiratory effect

Physiological Effects of Mineral Dusts

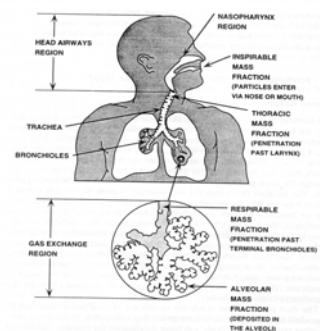


FIGURE 4.2 Graphic representation of the human respiratory system. (Modified after Lioy, 1995, from *Air Sampling Instruments*, 8th Ed., ACGIH, Cincinnati, OH. Adapted with permission.)

Respiratory system's clearance mechanisms:

Nose – hair, mucous

Mouth – mucous

Trachea, bronchi, bronchioles – mucous, cilia

Alveoli – surfactant, macrophages (phagocytes), lymphatic system

Terminology on inhaled particles:

Inspirable – capable of being inhaled into the nose or mouth; $< 100 \mu\text{m}$

Thoracic – reach past mouth and nasal region; $< 25 \mu\text{m}$

Tracheobronchial – $5\text{-}25 \mu\text{m}$

Respirable dust – enter alveolar region; generally $< 5 \mu\text{m}$; some up to $10 \mu\text{m}$

Respirable Dust Penetration Curves

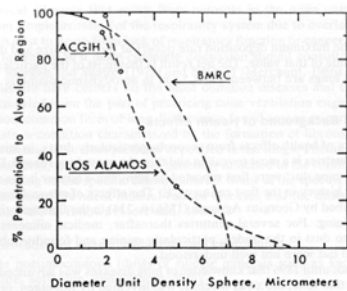


FIGURE 4.3 Characteristic curves for penetration of aerosols into the alveolar region of the lungs. [Copyright (1970), American Industrial Hygiene Association, reprinted with permission.]

Respirable Dust Penetration Curves

Based on aerodynamic diameter – diameter of particle with a density of 1 g/cm^3 that has the same aerodynamic properties as a given particle of arbitrary shape

Not all particles inhaled to the lungs are deposited; the maximum deposition rate occurs at $3 \mu\text{m}$

Respirable Dust Deposition Curves

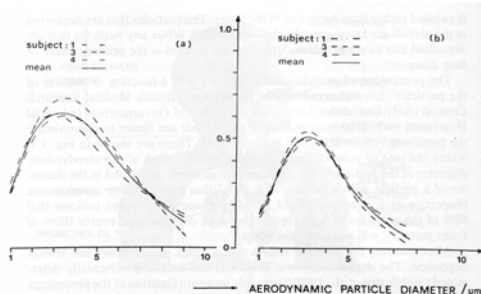


FIGURE 4.4 Deposition curve for aerosol particles in the alveolar region of the human lung for two breathing patterns: (a) breathing rate = $250 \text{ cm}^3/\text{s}$; (b) breathing rate = $750 \text{ cm}^3/\text{s}$. (For other variables of the breathing patterns, see original article.) (After Stahlhofen et al., 1980, copyright American Industrial Hygiene Association, used with permission.)

Historical Background of Health Effects

- Pliny the Younger – diseases first reported in first century
- Agricola – described health effects in 1500s
- For centuries, diseases recognized but not well understood
- 1896 Roentgen invented x-ray machine, which would be used later to diagnose disease

Historical Background of Health Effects

- Gauley Bridge, WV – galvanizing event in U.S. (1930s); protection laws vs. silicosis
- Extensive research on coal workers' pneumoconiosis just before WW II, then accelerated afterwards
- Asbestos link to cancer proved in the 1970s
- Research results from animal studies link DPM to cancer in 1980s

Respiratory Ailments and Diseases

- Silicosis
- Silicate pneumoconiosis
- Asbestosis
- Coal workers' pneumoconiosis
- Beryllium disease
- Siderosis

Dust Harmfulness Factors

- Composition
- Concentration
- Particle size
- Exposure time
- Individual susceptibility

Explosive Dusts

- When suspended in air in high concentration
- Large surface area of particles
- Powerful chemical reaction upon oxidation
- Ignite from open flame, electric arc, blast, methane ignition, autoignition at critical temperature

Explosive Dusts

- Explosions in air-methane mixtures propagate at speed of sound (1117 fps); called detonating explosions
- Air-dust mixtures propagate slower; flame speed ~ 30-35 fps; called deflagrating explosions; self-feed since shock wave travels faster, stirring up dust
- Extensive research on explosions

Coal Dust Explosions – Explosibility Variables

- Composition
- Particle size
- Concentration
- Flammable gas
- Moisture effects
- Presence of incombustible material
- Other dusts

Lean flammability limit for mixtures of methane and coal dust

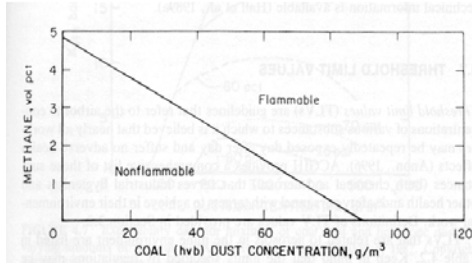


FIGURE 4.6 Lean flammability limit for mixtures of methane and coal dust. (After Hertzberg and Cashdollar, 1987. Copyright ASTM. Reprinted with permission.)

Explosibility data for mixtures of coal dust and limestone dust for varying amounts of rock dust

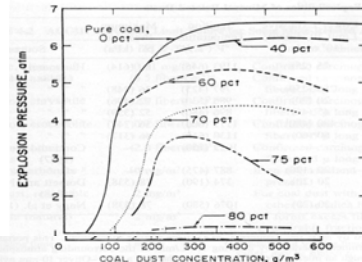


FIGURE 4.7 Explosibility data for mixtures of coal dust and limestone dust for varying amounts of rock dust. (After Hertzberg and Cashdollar, 1987. Copyright ASTM. Reprinted with permission.)

Explosibilities of mineral-related dusts

TABLE 4.1 Explosibilities of Mineral-Related Dusts

Dust	Lower Explosive Limit 10^{-3} oz/ft ³ (10 ³ mg/m ³)	Ignition Temperature °F (°C)	Maximum Pressure psi (kPa)	Source
Aluminum	25 (25)	1193 (645)	89 (614)	Hartmann (1948a, 1948b)
Iron	250 (250)	797 (425)	36 (248)	" " "
Magnesium	20 (20)	986 (530)	72 (496)	" " "
Titanium	45 (45)	896 (480)	52 (359)	" " "
Zinc	480 (480)	1112 (600)	36 (248)	" " "
Coal	60 (60)	1130 (610)	46 (317)	" " "
Sulfide ore	—	1022 (550)	—	Conti and Hertzberg (1987)
Oil Shale	120 (120)	887 (475)	—	" " "
Sulfur	20 (20)	374 (190)	78 (538)	Dorsett and Nagy (1968)
Gilsonite	20 (20)	1076 (580)	78 (538)	Nagy et al. (1965)

TABLE 4.2 ACGIH Threshold Limit Values for Some Mine-Related Aerosols

Substance	TLV-TWA	Comments
Arsenic, elemental	0.01 mg/m ³	Confirmed carcinogen
Asbestos, amosite	(0.5 fibers/cm ³) ^a	Confirmed carcinogen, applies to fibers >5 μ long
Asbestos, chrysotile	(2.0 fibers/cm ³) ^a	Confirmed carcinogen, applies to fibers >5 μ long
Asbestos, crocidolite	0.2 fibers/cm ³	Confirmed carcinogen, applies to fibers >5 μ long
Asbestos, other forms	(2.0 fibers/cm ³) ^a	Confirmed carcinogen, applies to fibers >5 μ long
Calcium carbonate	10 mg/m ³	If no asbestos and <1% silica present
Coal dust, respirable	2 mg/m ³	For coal dust with <5% silica; otherwise silica limit applies
Graphite (natural)	2 mg/m ³	All forms except fibers; respirable fraction only
Kaolin, respirable	2 mg/m ³	
Magnesite	10 mg/m ³	For total dust with no asbestos and <1% silica
Mica, respirable	3 mg/m ³	
Nuisance dusts	10 mg/m ³	For total dust with no asbestos and <1% silica
Oil mist, mineral	5 mg/m ³	Excludes vapor; TLV-STEL is 10 mg/m ³
Perlite	10 mg/m ³	For total dust with no asbestos and 1% silica
Portland cement	10 mg/m ³	For total dust with no asbestos and 1% silica

TABLE 4.3 Aerosol Sampling Instruments

Type	Collection Principle	Examples with Mine Application	Additional Information
Elutriators	Gravitational separation of larger particles from smaller particles	Hexhlet horizontal elutriator MRE gravimetric dust sampler	Hering (1989, p. 381) " " "
Cyclones	Centrifugal separation of larger particles from smaller particles	Dorr-Oliver 10-mm cyclone	Hering (1989, pp. 379–380)
Impactors	Impaction of flowing particles on a solid surface	Anderson/Maple Model 298 multistage impactor MOUDI (microorifice uniform deposit impactor)	Hering (1989, pp. 373–374) Hering (1989, p. 372)
Impingers	Impingement of particles into a liquid	Greenburg-Smith impinger Midget impinger	Hering (1989, p. 382) " " "

Table 4.3 Aerosol Sampling Instruments (Continued)

Filters	Combination of inertial impaction, interception, diffusion, electrostatic, attraction, and gravitational forces	Fiber filters	Lippman (1989, pp. 305–336)
		Cellulose	" " "
		Glass	" " "
		Paper	" " "
		Membrane	" " "
		Millipore	" " "
		Nucleopore	" " "
		Silver	" " "
		Polyvinyl chloride	" " "
Electrostatic precipitators	Electrical charging of particles with collection on electrode of opposite polarity	Seldom used in mines	Swift and Lippman (1989, pp. 387–403)
Thermal	Particle movement toward a cooler surface in a thermal gradient	Seldom used in mines	" " "

Source: Modified after Glenn and Craft (1986) and First (1989).

Measures of Aerosol Exposure

$$\text{Dose} = \sum_{i=1}^n E_i T_i \quad (\text{integral form, too})$$

E_i – exposure

T_i – time period

Measures of Aerosol Exposure

$$\text{Deposited dose} = IP_p P_d \sum_{i=1}^n E_i T_i$$

I – inhalation rate, m³/h

P_p – probability that measured dust will penetrate to alveoli

P_d – probability that dust penetrating alveoli will be deposited

Measures of Aerosol Exposure

Lastly,

$$E_{\text{TWA}} = \text{dose/total exposure time}$$

Diesel Particulate Matter (DPM)

- Fumes and soot from incomplete combustion
- Spherical carbon core < 1 μm
- Polycyclic aromatic hydrocarbons (PAHs) and other chemical compounds are absorbed on carbon core
- Particles agglomerate and form clusters

Measurement of DPM

- Many studies using 10-stage impactors, MOUDI (gravimetric)
- Complications in analysis (see figure, next slide)
- Dichotomous sampler developed
- Respirable Combustible Dust method
- NIOSH Method 5040 – carbon analysis => elemental carbon

In-mine DPM samples

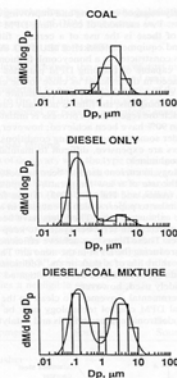


FIGURE 4.9 Size distributions of coal, diesel, and mixed aerosols measured under laboratory conditions. (Cantrell and Rubow, 1991, used with permission.)

**Sources of Dusts
in Mines
Discussion**

**Next Lecture
Aerosol Control
Technology and
Personal Protective
Equipment**